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# The natural history and fisheries ecology of Lake Chilwa, southern Malawi

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ABSTRACT

Lake Chilwa produces between zero and 24,000 metric tons of fish per year, making it one of the most productive but variable lakes in Africa. The size of the lake varies seasonally and among years, sometimes drying completely. Its surrounding wetland and floodplain provide habitat for a diversity of birds and economically valuable grasses and reeds. When the lake has water, there is considerable activity on its shores and temporary fishing villages spring up. People move in and out of the lake basin in concert with these seasonal and longer term changes. This paper examines the environmental dynamics of Lake Chilwa and its surrounding wetlands, presents an overview of the socio-economic context of the area and discusses threats to this resilient system that might occur as a result of climate change. We conclude that management of Lake Chilwa must place the lake in the wider economic and ecological system in which it is situated. Ultimately, land-use practices within the basin present more of a threat to the resilience of the fishery and people's livelihoods than overfishing or a strict focus on the lake's resources. These perspectives present significant challenges to conventional fisheries governance.

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## Introduction

The shallow lakes of sub-Saharan Africa are among the most productive aquatic ecosystems in the world (Talling and Lemoalle, 1998). These lakes are famously variable, and some even dry

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completely in low-rainfall years. The natural ecosystems of Africa's shallow lakes and the people who depend on them appear to be well adapted to these cycles of change (Sarch and Allison, 2000; Jul-Larsen et al., 2003). Arguably these attributes make the ecology of these linked human–environment systems pre-adapted to coping with human-induced change (Moss, 1992). Such resilience is not often emphasized in fisheries analyses, typically pre-occupied with stability as a management objective. Here we analyze the nature of climate and fishery variability, livelihood strategies of people living around Lake Chilwa in southern Malawi, and attempts to manage the lake's fishery.

In 2003, Eyolf Jul-Larsen and coauthors published a comprehensive and provocative analysis of the management of freshwater fisheries in southern Africa, including Lake Chilwa (Jul-Larsen et al., 2003). Their conclusions profoundly challenged prevailing paradigms of target resource oriented management and the context-free development of co-management institutions. In their view, management measures that sought to control the number and size of fish caught, and where that fishing happened, had failed. So too had governance arrangements that appeared more as instruments of government policy and traditional authority than genuine attempts to devolve power and allow for more equitable flow of benefits.

The analysis provided by Jul-Larsen et al. (2003) built on earlier challenges to prevailing views of how humans interact with natural resources in the region. On a broad scale, concepts drawn from 'new ecology', with its emphasis on non-equilibrial dynamics and strong bottom-up biophysical forcing of certain types of ecosystems (e.g. semi-arid rangelands, shallow lakes, upwelling systems) provided an intellectual frame for discussion (Botkin, 1990; Behnke et al., 1993; Scoones, 1995). Likewise, in fisheries, ideas from chaos and complexity theory suggested that regulating fish catches on the basis of optimal yield models was untenable (Wilson et al., 1994). On a smaller scale, the monograph by Margaret Kalk and co-authors (Kalk et al., 1979) on the ecology of the Chilwa basin came to many of the same conclusions and provided a classic early example of an interdisciplinary analysis of a fishery system. Looking back, Botkin, Behnke et al., Scoones, Jul Larsen et al. and, 35 years before them, Kalk et al., presaged many concepts now gathered and developed under the banner of 'resilience thinking' (Berkes et al., 2003; Folke, 2006).

In this review we update the analyses provided by Kalk et al. (1979) and Jul-Larsen et al. (2003) with particular reference to the vulnerability of fisherfolk in the Lake Chilwa Basin to climate change and their capacity to adapt to this potentially dominant external driver of ecology and social dynamics. We begin by providing an overview of the environmental and socio-economic context of the Lake, then provide further detail on the Lake Chilwa ecosystem, and continue with an historical overview of the development of management institutions in the fishery. Finally, we conclude with some thoughts about building climate change adaptations in such a dynamic system.

## The Lake Chilwa Basin—environmental and social context

The Lake Chilwa Basin (Fig. 1) lies in a tectonic depression in south-eastern Malawi (latitude 15° 30" S and longitude 35° 30" E). Rain falls during November to March only, followed by a very long dry season when virtually no rain falls and no crops can be grown except under irrigation. Demand for irrigation water is also increasing due to winter cropping and wetlands are being converted to rice cultivation. Siltation of rivers has degraded water quality and resulted in flooding in areas throughout the basin. The rivers in the basin supply 5.75 million cubic meters of potable water annually valued at US $2.5 million to urban and peri-urban areas of Zomba, Namadzi and Domasi (SRWB, 2009). The basin also supports a large number of nationally and internationally important flora and fauna. For example Lake Chilwa and its surrounding wetlands and floodplains sustain about 153 species of resident and 30 species of palearctic (migratory) birds, respectively. The varied habitats of Lake Chilwa and surrounding wetland provide food, shelter and breeding areas for the waterfowl, which are vulnerable to overexploitation due to high demand and pervasive poverty.

Just over 200 years ago Lake Chilwa was 9 meters higher than its present level, and therefore extended over a much larger area than its present size (Crossley et al., 1984). There is much evidence from terraces and sandy raised beaches and sand-spits of much higher lake levels (up to 33.5 m) above the present lake level, and this has determined the nature of large areas of the Chilwa plain. The lacustrine and alluvial deposits which occupy the greater part of the Lake Chilwa area are other indicators of its earlier size (Garson, 1960). To the north Lake Chilwa was continuous with Lake Chiuta, from which it is now separated by a sandbar, formed about 8000 years ago, and to the south it extended to near the western foot of Mulanje Mountain, where possibly there was an outlet into the Ruo river valley.

The Lake Chilwa catchment is bounded in the west by the Chikala Hills, Zomba and Malosa Mountains, the Shire Highlands and Chiradzulu Mountain, which give rise to the Zumulu, Lingoni, Domasi, Songani, Naisi, Mulunguzi, Likangala, Thondwe and the Namadzi and Mombezi rivers, which join the Phalombe river (Fig. 1). The catchment is bounded to the south-east by Mulanje and Mchesi Mountain, which give rise to the Phalombe River and the Sombani River, which forms Mpoto Lagoon before flowing into the lake. To the east lie the hills and mountains of Mozambique, which give rise to the Bungwe, Mnembo, Matchimaze, Namajete and Cocole rivers. In the north-east the Mikoko and Naminga rivers rise from the eastern rift.

Most of these rivers have a very high peak flow in the months of February to March, as much as 200 times greater than the very low base or minimum flow over the rest of the year (EAD, 2000). These patterns in rainfall are reflected in lake levels, which are highly correlated with annual flow in rivers and streams and precipitation in the catchment (Delamore, 1987). This association between rainfall and the lake level can even be seen with the rainfall four years before, the greatest effect being from the rainfall in the previous year. It indicates that the hydrology of the lake basin has a number of flow-dampening components such as wetlands in the mountain source areas, some groundwater infiltration and recharge to surface flows in the upper and mid reaches of the rivers, and hydrological storage in surrounding lake edge wetlands. The large rivers converging into the swampy and deltaic land at the lakeshore create pools ranging from 3.6 to 5.1 m deep (J. Wilson, unpublished data) which are the key to the survival of residual fish stocks when the lake dries, and the rapid recovery of the fishery following the refilling of the lake (Ambali and Kabwazi, 1999). It is these deep pools, which make Lake Chilwa unique, and which are under threat from siltation by nearby farms.

Lake Chilwa's biologically diverse wetlands were classified as a Ramsar site in 1997. They provide economic benefits to the people who live there but these resources are under pressure. Degradation in the upper catchment including deforestation, poor agricultural practices, lack of soil erosion control and steep slope cultivation affect the livelihoods of all catchment residents. These practices lead to increased silt loads in the rivers during heavy rains, which in turn lead to further loss of topsoil and a decline in agricultural yields. Downstream flooding, particularly in the areas of the flood plain rice schemes, results in the destruction of houses and sometimes forces relocation of entire villages (J. Wilson, personal observation).

The Lake Chilwa Basin is one of the most densely populated areas in Malawi, and indeed in Africa, with roughly 162 people per square kilometer. The population of Lake Chilwa basin is predominantly rural and 60% of the population depends on farming for their livelihoods (Zimba and Kaunda, 1999). Mean household incomes are low and range from 5000 to 25000 Malawi Kwacha a year (USD113–568 per year at 1999 exchange rate) and literacy is poor (less than 50% of men and 30% of women are literate; Binauli and Chipeta, 1999; EAD, 2000).

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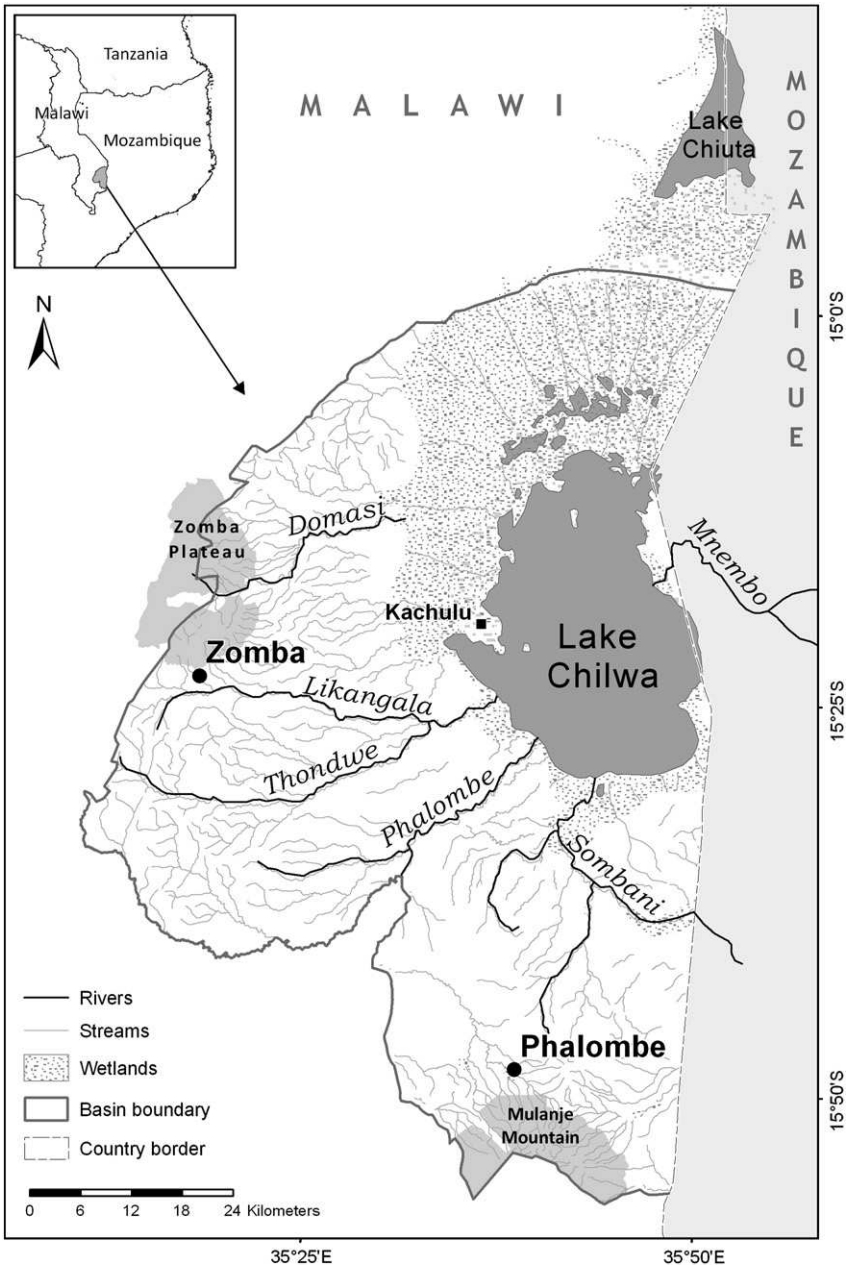


**Fig. 1.** Map of Lake Chilwa and surrounding basin within Malawi. In addition to major towns and rivers (in italics), the location of Kachulu, a major landing site and location of the lake level gauge is shown.

High population density, together with the fact that 44% of households have less than 1 hectare of land on which to subsist, mean that the people of Lake Chilwa are among the poorest in Africa. The yields of the main subsistence crops are very small, ranging from 1000 to 1250 kg $ha^{-1}$ in Zomba District falling to less than 1,000 kg $ha^{-1}$ in Machinga District east, but rising to 1,250–1,375 kg $ha^{-1}$ in Phalombe district (EAD, 2000). Most households (85%) rely on food from their own farms and this is only sufficient to feed the family for about half the year. At the time of the Malawi Core Welfare Indicators survey, from September to November 2002, 93% of all households had run out of food (GoM, 2002).

In Malawi, the poverty line is based on the cost of the minimum daily calorie requirements plus some additional basic nonfood items, which is about 41 U.S. cents (58 Malawi Kwacha at 2009 exchange rate) per person per day. The ultra-poverty line is estimated at 25 U.S. cents (35.5 Malawi Kwacha). Based on the poverty mapping analysis, 64.3% of Malawi's population live in poor households. Around Lake Chilwa an even greater proportion of the population (70–80%) live below the poverty line. Furthermore, 36% of Malawi's population lives in ultra-poor households and in the districts around Lake Chilwa, the range is from 36 to 50% of households (GoM, 2002; GoM and IFPRI, 2002).

In a comparison of data on livelihood strategies and options between the 1960s and early 2000s, some changes emerge. There has been a decline in livestock holdings and growth in rice production. In addition, forest cover has decreased significantly and cultivation has increased throughout the basin, including up to the edge of river banks, a practice that greatly increases erosion. In other areas, patterns remained similar such as in the wealth differentials between farmers and fishers. With the exception of times when the lake dried completely, the incomes of fishers are higher than for farmers in the basin. Similar to results from studies carried out in the 1960s, more recent data show differentiation among those who fish, with a small minority who own boats and gear earning larger incomes. Surveys

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carried out on Lake Chilwa reveal the diversity of strategies among fishers (Allison and Mvula, 2002). There are migrant fishers and fish traders who come in and out of the lake throughout the year. They are highly mobile and the fishers may come alone or bring their wives and families with them to temporary camps. In these migrant groups, fishing and fish trading contribute over 80% of their household income.

Residents who live in villages near the lake engage in a mixture of fishing (for petty trade and subsistence), wage-labor, farming and self-employment. For this group, fishing may contribute 30–50% of their household income. Generally, households which engage in fishing have higher incomes than those who do not. Ellis et al. (2003) demonstrated that involvement in fishing and other off-farm activities typified the better off households. Many of those engaged in fishing in Lake Chilwa are migrants and their living conditions are far from ideal. Many live in camps with no sanitation facilities and away from access to health facilities or security. Furthermore, they have high exposure to malaria and bilharzia and are at great risk of HIV/AIDs exposure. So, while fishers often earn higher incomes, they are also exposed to greater risks and hazards.

The Lake Chilwa wetland is extremely important for the economy of the entire basin, with fish contributing up to USD17 million a year in net benefits, grasslands providing USD589,500, open water USD402,557, agricultural land USD385,212 and vegetation USD12,434 (Schuijt, 1999). Even though the fishery of Lake Chilwa offers an economically unstable environment, determined by seasonal and long-term fluctuations in the lake level, at high production periods fisheries contribute significantly to the wider rural economy. According to Schuijt (1999), the value of fish is ten times higher than rice, the second most important economic activity in the basin and, per square km, fishing provides greater economic returns than agriculture or livestock grazing. Cash earned through fishing or trading is invested in farming (purchasing of inputs), and sometimes in other small businesses.

A whole array of services spring up around the landing sites from room rentals for migrant fishers to shops selling goods and restaurants selling food and beer. Many people sell small amounts of goods as well in these areas (Allison and Mvula, 2002). Thus, the fishery is the engine of the basin's economy. The most recent drought in 1995 in which the lake dried up completely had a devastating effect on the local economy. As people withdrew from fishing, greater pressure was put on other basin resources, from birds to trees, to meet the need for food and cash income. Both for sustainable livelihoods and biodiversity, efforts must be directed at creating and maintaining the environmental conditions necessary to sustain the productive potential of the fishery as much as is possible in such a fluctuating system (Kalk et al., 1979; Allison and Mvula, 2002).

## The natural history of Lake Chilwa

Lake Chilwa is shallow, the average depth ranges between 1.5 and 2.5 m, with a maximum of 5 m in the south-east. The lake and wetland together are about 40 km across and 60 km in length. Open water can cover 1500 km$^2$ in years of 'normal' rainfall (EAD, 2000). The lake is endorheic—fed by many rivers and streams but with no surface outlet. The depth of the lake varies significantly among years, sometimes changing more 4 m in a single year (Fig. 2). The lake dries periodically (nine times in the 20th Century, at 10- to 20-year intervals). The drying of the lake, while being temporarily disastrous for the fishery, greatly enhances the lake's productivity, as evidenced in the increased fish catches, nearly 23,000 tons in 1979 from a mean of 10–13,000 tons p.a. five years after the recovery from drying in 1973 (Kabwazi and Wilson, 1998). The drying of the lake allows for the oxidation and release of nutrients from the very large quantities of organic matter, largely derived from the surrounding *Typha domingensis*, the False Bullrush, swamps that settle on the lake bed.

Because the lake has no outlet, its centre is slightly saline with an outer ring of freshwater at its margins. Salinity normally varies from 1 to 1.5 ppt, increasing to 10–12 ppt in years of minor recession (EAD, 2000). This salinity, combined with a depth of 1 meter or less around much of the lake, especially in the north, results in *Typha* forming a dense swamp around the margin of the lake. This is in contrast with other shallow African lakes and rivers, where Papyrus (*Cyperus papyrus*) dominates (Howard-Williams and Walker, 1974). In Lake Chilwa, Papyrus is found only in small stands in the mouths of inflowing rivers.

The natural history of the lake is governed by four major spatial characteristics and two predominant time scales of variability (Kalk et al., 1979). The spatial characteristics are: (i) the Chilwa-Chiuta sandbar that closed off the Chilwa drainage basin so that it became one with no outflow; (ii) the shallowness of the lake (mean depth *ca.* <2 m); (iii) the dominance of the *Typha* swamp which covers half the flooded area of the lake and is ringed with an almost equal area of seasonally inundated floodplain and (iv) the existence of deep pools in the inflowing rivers which serve as critical fish refuges during lake drying periods. The two predominant timescales governing ecosystem variability are:

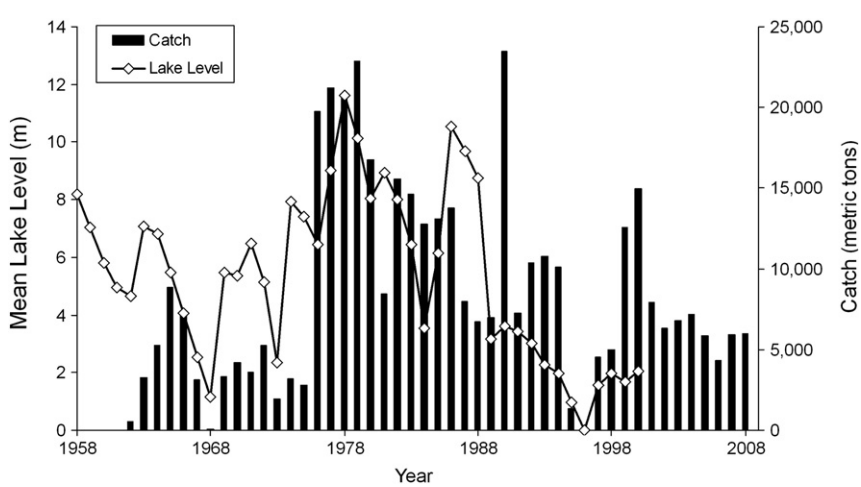


**Fig. 2.** Historical (1949–2000) mean annual lake level (m) of Lake Chilwa as measured at Kachulu and total catch (metric tons) from the lake between 1962 and 2008. Catches were smallest in 1995 and 1996 due to lake recession and a fishing ban.
Sources: lake levels from GoM (2003) and catches from GoM (2005; GoM unpublished data).

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(a) Seasonal and (b) Inter-decadal. The ecology of the lake is discussed in relation to these time and space characteristics.

*Living in a closed drainage basin*

While plants and most animals (insects, birds, mammals) are not restricted by closure of the lake system, it is obvious that migratory fish are affected. There are 36 species of fish in Lakes Chilwa and Chiuta, five of which occur in Lake Chiuta but not Lake Chilwa. It is not known whether these species arrived in Lake Chiuta after it became separated or entered Lake Chilwa during high water periods, but could not survive the more variable conditions after the sand bar formed (Kirk, 1976).

The principal feature of closed drainage basins is their sensitivity to climate because the difference between inflows to the lake (driven by the precipitation on the lake and catchment) and evaporation from the lake is precisely mirrored by lake level change. When evaporation exceeds inflow the lake level drops and *vice versa*. Closed drainage lakes are therefore characterized by considerable variability in lake levels and the Lake Chilwa level record shows an annual halving of lake volume with an annual variability in lake level of ca. 0.8–1.2 m in a so-called 'normal' year and an almost complete drying up of the lake at inter-decadal (10- to 20-year) time periods (EAD, 2000).

When Lake Chilwa levels fall, the floodplain dries up, parts of the surrounding swamp may dry up and the lake shrinks; the salt concentration increases proportionately to the decrease in volume and the ion ratios change as some salts are selectively precipitated (Morgan and Kalk, 1970). In addition, wind-induced mixing of bottom sediments into the shallow lake water increases, and deoxygenation in the swamps (with decreasing water volume) and the open lake (with sediment having a high oxygen demand being stirred into the water column) becomes more prevalent. These environmental changes are caused by lake level changes of only a meter or less and all affect the lake ecosystem (Kalk et al., 1979).

As with all closed drainage basins, salts have accumulated over time and Lake Chilwa is characteristically rich in dissolved salts with total salt concentrations of 1.5 ppt (wet season) to 2.5 ppt (dry season), some 100 times higher than Lake Malawi. However, despite this salt richness, the open lake in a 'normal' year still falls within the 'freshwater' limit of 5 ppt of salt suggested by Beadle (1974). This is a point on a continuum from strictly freshwater to strictly saline water which, in general, separates those organisms with a hypotonic physiology from those that do not. In a normal year the organisms inhabiting Lake Chilwa are therefore not in any physiological stress from changing salt concentrations and the lake has a predominantly freshwater fauna and flora.

*Living in a shallow lake*

Shallow lakes such as Lake Chilwa have distinctive features that make them highly productive. At the same time, they are obviously very sensitive to any catchment management approaches that are likely to alter water levels and inflows. Howard-Williams and Ganf (1981) outline the following key features of African shallow lakes: a high surface area to volume ratio; significant influence of wind mixing on daily and seasonal lake structure and oxygen regime; and a high degree of sediment-water interaction that alters the turbidity of the water, the salt and nutrient concentrations and the silt content.

A large surface area to volume ratio results in a reduced ability of a lake to buffer solar heating and so Lake Chilwa has warm waters that may vary considerably between day and night. A 6.2 °C change between day and night was recorded in January 1970 and there may be times when even greater temperature excursions occur (Moss, 1979).

The waters of Lake Chilwa are mixed by winds every day and are often calm at night. This has the effect of circulating algae in the water column, allowing photosynthesis even in the turbid conditions and ensuring adequate oxygen for respiration before nightfall. Winds blow oxygenated lake waters into the surrounding swamps in the dry season and even across parts of the northern grasslands in what are termed "wind tides," and on their return to the open lake these waters transport vegetative matter back to the lake. Wind tides also deposit suspended sediment in the swamps, particularly in the north.

The bottom sediments of Lake Chilwa are continually stirred in to the water column so the lake is continually highly turbid and the resulting re-suspension and deposition of silt and clay particles makes the lake bed unfavorable for many invertebrates and the waters unfavorable for many predacious bird and fish species. The silt itself is a stressor, clogging animal filtering mechanisms and fish gill membranes. Nutrients bound on sediment particles, such as phosphorus, are readily exchanged with lake water in a suspended state (Moss, 1979).

*A swamp-lake integrated ecosystem*

A characteristic feature of Lake Chilwa is the very large area of surrounding swamp and flood plain that imparts special attributes on the functioning of the lake. The open lake margin is characterized by a dense wall of *Typha domingensis* which occupies the shallow margins of the lake and covers an area of 500–600 $km^2$ (van Zegeren, 1998). This merges on the landward side into a very diverse floodplain of marshes, where the perennial rivers flow, and seasonally inundated floodplain grasslands between the rivers.

The wetlands of Lake Chilwa provide a very large area of suitable habitat for fish, invertebrates, zooplankton, algal species, birds, insects and reptiles that otherwise do not occupy the open lake. The vegetation also acts as a trap and a 'sieve' for incoming sediments from the catchment, and provides a very significant source of organic matter to the lake system (800,000 metric tons of dry matter per year from *Typha* alone; Howard-Williams, 1979). The decomposing material from the swamps and marshes is a substrate for mud-dwelling invertebrates of the open lake and a food source for zooplankton, fish and many bird species (Kalk et al., 1979). These functions vary seasonally with changing lake levels and as waters move through the swamps to the lake in the wet season. The Lake Chilwa ecosystem is therefore an integrated unit of open lake water, swamp and floodplain that gives it a special characteristic and assists in buffering the lake from the impacts of a highly variable water level regime as is discussed below.

*Temporal change 1: Seasonal time scales (a "normal" year)*

At latitude 15° S there is a strong seasonality that is not seen in shallow lakes closer to the equator. This seasonality is driven by the contrast between a marked wet season (November to March) when the Inter-tropical Convergence Zone moves down through southern Malawi, and the dry season from May to September when southern Malawi falls under the Southern African Anticyclone. April and October are months of change between these predominant weather systems. The annual variability that follows from the wet and dry seasons controls much of the lake's ecology because it controls lake levels and the changes associated with these different levels.

Each year during the mid-rainy season (December–March) the river inflows approach their maxima and these spread out through the swamps and, together with rain on the flood plain itself, result in a flooded area of *ca.* 1200 $km^2$. In December, when lake levels are still low, these floods move across the plains and through the swamps displacing swamp waters out into the lake and resulting in greatly reduced salinities in the swamps and to a lesser degree around the periphery of the open lake.

The seasons dictate the breeding cycles of the three most economically important resident fish species in the lake. Matemba (*Barbus* spp.) breed through the months of the rainy season. This species is noted for a high natural fecundity and a high intrinsic rate of natural increase. High fishing catch rates near the swamp margins lead to the assumption that the fish are intercepted during their breeding migrations into and out of the swamps and rivers. Although

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no direct evidence of egg laying is available, the swamps provide a substrate for egg mass and protection for the newly hatched fry (Furse et al., 1979a).

For Mlamba (*Clarias gariepinus*), the main breeding season appears to be from November to January. Breeding behavior involves characteristic migrations from the lake into the surrounding swamps, rivers and floodplains to spawn. Makumba (*Oreochromis shiranus chilwae*) have a long breeding season from September to May with females likely to breed 2 or three times in a year. They require sandy substrates to nest in before mouth brooding and these are mostly found on the eastern side with pockets on the north and west of the lake. This species does not leave the lake to spawn. All of these fish are well-adapted to the highly variable conditions of the lake and indeed their breeding habits provide them with the ability to bounce back after suffering severe drying periods, when they take refuge in the deep pools in the inflowing rivers.

*Temporal change 2: inter-decadal timescales (a dry year and its recovery)*

Zooplankton respond to the drying phase of the lake in 1966 by producing robust "resting stages" such as ephippia (thick shelled resting eggs) or by going into diapause and aestivating in the dry mud (in the case of the crustacean *Mesocyclops*). Kalk (1979a) reported that diapause in zooplankton was initiated when the conductivity of the lake exceeded 5000 μS $cm^{-1}$, the then shallow water was supersaturated with oxygen and pH rose above 9.0 with afternoon temperatures exceeding 35 °C. During the year of refilling, rotifers recovered more rapidly than crustaceans, due to their shorter hatching time and quicker generational turnover time. The subsequent rapid recovery of crustaceans after refilling showed *Mesocyclops* as the first to appear as adults which, in turn, fed on the increasing rotifer populations. Kalk (1979a,b) reported that in the first year after refilling, *Clarias* (Mlamba) fed extensively on the recovering zooplankton, with *Barbus* spp. preying on zooplankton in the second year of refilling. Fish predation was assumed to be the reason for the slow recovery of zooplankton to full post-recovery levels.

Benthic invertebrate populations crashed during the drying of the lake in 1968–1969, particularly when the main body of the lake and the swamps became separated. As a result, the benthic habitat provided by the swamp vegetation was no longer available to invertebrates in the lake. The chironomid midge *Nilodorum brevibucca* was the first to disappear followed by the water boatman (*Corixa* sp) and then the snail (*Lanistes ovum*), whose shells littered the lake bed as far as was visible from the shore (MacLachlan, 1979). Benthic invertebrates rapidly colonized the refilling lake. Insects flew in from adjacent water bodies and other animals such as molluscs survived by aestivating in the mud.

The dry periods do not appear to greatly influence bird populations, and even the predator bird species of the open lake such as the Pink Backed Pelican (*Pelecanus rufescens*), Grey Headed Gull (*Larus cirrocephalus*) and the terns returned to the lake within the first year of the its refilling after the 1968 dry period (van Zegeren and Wilson, 1997a, 1997b). Many of the marsh birds, ducks and geese also returned during lake refilling because their refuge habitats during dry periods at Lake Chilwa were easily accessible both to the North (Lake Chiuta basin, upper Shire River) and to the South (wetlands of the Lower Shire River). However, with the loss of the fishery, local people turn to waterbirds as a source of food and income, and in 1996, an estimated 1.2 million waterbirds were trapped (snared) or shot following the drying of the lake in 1995 (Van Zegeren and Wilson 1997a,b).

*Aquatic invertebrates*

While there are at least 10 species of zooplankton in Lake Chilwa, only three large (1.0–1.3 mm) crustaceans (*Diaphanosoma excisum*, *Tropodiaptomus cunningtoni* and *Daphnia barbata*) are abundant (Kalk, 1979a,b). Smaller crustaceans are also present, with one (*Alona* sp.) being associated with aquatic plants of the lake edge. Three species of rotifers may be found in the rainy season when lake levels were rising but they are not numerous. Although Lake Chilwa has a lower zooplankton diversity than many other African lakes, densities are high (100–800 $l^{-1}$) with maximum densities recorded near the swamp lake edge and during the rainy season. All species are tolerant of salinity changes during a normal year (Morgan and Kalk, 1970; Kalk, 1979a,b).

The marked changes in physico-chemical conditions imposed by seasonal, and particularly inter-decadal variability, has imposed a set of rigorous conditions that few benthic invertebrates are able to tolerate with only 45 species recorded when the lake was in a 'normal' phase and 11 species during the drying phase when the lake became separated from the swamp. During a normal year Chironomidae (mostly *Nilodorum* spp.) comprised 63% of the benthic fauna (dry weight basis), with other insects (Diptera, Trichoptera, Hemiptera) making up 18%, molluscs (the snails *Lanistes ovum, Bulinus globosus* and *Biomphalaria* sp.—the latter two being vectors for bilharzia) 10%, and annelids 3% (MacLachlan, 1979). Aquatic invertebrates support a significant amount of fish production in Lake Chilwa as they are a major food source for *Barbus* sp. and *Clarias gariepinus* (Kirk, 1967).

The open lake sediments are virtually devoid of benthic invertebrates because the very fine soft sediments of the lake bed, continually re-entrained by wind action, are not a suitable habitat. The principal feature that controls benthic fauna in the open lake is the proximity to the swamp. The persistence of animals close to the swamp edge is due to plant debris from the swamps providing stable microsites for colonization. The swamp itself is a major habitat for aquatic invertebrates. Thus the swamp-lake margin is a critical habitat that influences lake productivity.

*The fishes*

Possibly as a result of the extreme conditions, Lake Chilwa is low in fish species diversity and contains only fourteen species. Fish families existing in Lake Chilwa are Alestiidae [*Brycinus imberi* (Peters, 1852) and *Hemigrammopetersius barnardi* (Herre, 1936)], Cichilidae [*Pseudocrenilabrus philander* philander (Weber, 1897), *Oreochromis shiranus chilwae* (Trewavas, 1966), *Tilapia rendalli* (Boulenger, 1897) and *Haplochromis callipterus* (Günther, 1894)], Clariidae [*Clarias gariepinus* (Burchell, 1822) and *Clarias theodorae* Weber, 1897], Cyprinidae [*Barbus paludinosus* Peters, 1852, *B. trimaculatus* Peters, 1852 and *Labeo cylindricus* Peters, 1862], Mormyridae [*Marcusienis macrolepidotus* (Peters, 1852) and *Petrocephalus catastoma* (Gunther, 1866)] and Schilbeidae [*Pareutropius longifilis* (Steindachner, 1914)]. Tilapias (*Oreochromis shiranus chilwae*), clariids (*Clarias gariepinus and C. theodore*) and small barbs (cyprinids of the genus *Barbus*) are prominent in the lake and in its tributaries because they can withstand harsh conditions by developing drought-resistant strategies (Msiska, 2001). Fish exploit Lake Chilwa's rivers and swamps, including its major river inflows, as spawning areas and refugia during periods of lake desiccation (Jamu and Brummett, 1999). In Lake Chilwa and other African rivers, *Barbus* species migrate up tributaries in shoals during the flood season to breed (Bourn, 1974; Bell-Cross and Minshull, 1988; Jamu and Brummett, 1999).

## The Fishery

*Barbus paludinosus* and *B. trimaculatus* dominate the fishery, accounting for more than 70% of the total catch (EAD, 2000). The Lake Chilwa fishery, and the *Barbus* component in particular, has been resilient to cyclical drying of the lake, overfishing and use of illegal gears. Records show that reduced catches and increased fish mortalities occur during even minor or moderate recessions (Fig. 2;

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Kalk et al., 1979; Njaya et al., 1996). In the past, minor recessions in lake level, sufficient to reduce fishing for one or two years, could be expected every six years or so. Major recessions that interfere with fishing in the open lake for 3–5 years could be expected every 10–20 years (op. cit.; Lancaster, 1979). When the lake recedes, the fishery collapses but rapidly recovers within three to four years after the water returns. For example, after the 1995 recession, fish production from the lake reached about 12,500 metric tons within three years after the lake re-flooded (Fig. 2; Kabwazi and Wilson, 1998).

Between 1962 and 2008, the average annual fish catch of Lake Chilwa was 8,603 metric tons, but varied enormously among years, from catching very few fish when the lake dried in 1996 to an annual production of 23,485 tons registered in 1990 (Fig. 2). During the recession period (1995–1996), fishing operations were either suspended or reduced to a lower scale in river mouths. Although there is a clear and statistically significant correlation between lake level and catch ($r_{(37)} = 0.43$, $p < 0.01$), interpretation of trends is complicated by a range of factors. Firstly, the relationship between lake level and lake area is non-linear—as the lake fills past a threshold level (at around 4–5 m) the water rapidly spreads into the wetlands, increasing the area of suitable habitat for fish enormously. Conversely, as the lake recedes water is concentrated in the main body of the lake, greatly increasing the catchability of fish. This last phenomenon, along with the consequent increase in fishing effort (D. Jamu, personal observation) provides partial explanation for the apparently anomalously high catches in 1990 and in 1999–2000 (Figs. 2 and 3). Other complicating factors include problems in data collection from a dispersed and highly dynamic fishery, and the complex interactions between catch, lake level and fishing effort (see discussion of effort below).

The fish catches recorded for Lake Chilwa make it one of the most productive freshwater lakes in Africa. The lake's productivity of 159 kg $ha^{-1}$ (Furse et al., 1979b) is higher than that of lakes Tanganyika (90 kg $ha^{-1}$), Malawi (30–40 kg $ha^{-1}$) and Victoria (116 kg $ha^{-1}$) (Vanden Bossche and Bernacsek, 1990). The contribution of fish landings from Lake Chilwa to the national production levels varies on an annual basis with an average of 18% from 1993 to 2003. After the previous recession, the greatest contribution to national fish supply was 27%, in 1999 but this had dropped to 16% in 2003.

Based on water level and fishery changes, Lake Chilwa fish production follows three distinct phases (Kalk et al., 1979; Njaya, 2002). Firstly, the *normal* fishing phase, occurs between 2 and 3 years before or after major recessions. During this time, water level changes are on a seasonal basis. Secondly, the *drying phase*, which typically occurs at least 2 years before a major recession. During the drying phase, *Oreochromis* catches decline severely while *Barbus* catches initially increase but then drop in the last year of the drying phase. *Clarias* fishing stops in the last year of the drying phase. During this phase, remnant fish stocks find refuge in the mouths of larger inflowing rivers and in deeper lagoons. Thirdly, the *refilling phase* which takes place in the 2 years immediately after a major recession. In the refilling phase remnant fish stocks in river mouths and lagoons serve as a nucleus for natural restocking of the lake. During the major recession of 1968 *Clarias* populations recovered in 2 years, *Barbus* in 3 years and *Oreochromis* after 4- 5 years (Furse et al., 1979b).

Lake Chilwa fishers use fish traps, gillnets, seine nets, and long lines. However, fish traps are common because they are less expensive as they are made of local materials compared to other gear types that need synthetic materials. Fish traps and seines largely exploit Matemba (*Barbus* spp.) while gillnets and long lines target large fish species such as Makumba (*Oreochromis shiranus chilwae*) and Mlamba (*Clarias gariepinus*). Since 1983, the highest number of gillnets (about 9000) were registered in 1988, 2000 and 2002. The largest number of seines was 738, registered in 1998 after the 1995/96 recession while fish traps were highest (33,000) in 1992 before the recession. Fishers operating long lines are common in the southern part of the lake, which is deeper. The introduction since the early 2000s of the *Usodzi wa Mululu* fishing method, involving the use of introduced non-selective 'gauze' wire seines, has increased the vulnerability of fish to exploitation because it requires clearing of aquatic vegetation.

There was a major increase in fishing effort around the early 1970s as the region became better integrated into the market economy and money was to be made selling fish. Since the 1995–1996 recession, the highest recorded numbers of gear owners and crew were 5,406 and 6,250, respectively recorded in 1998 (Fig. 3). However, the numbers decreased to about 2,400 crew and 2,700 gear owners in 2002 (GoM, 2005) mainly due to migration of fishers out of the fishery. Schuijt (1999) estimated that Lake Chilwa had 3,720 fish traders, 17 boat builders and "a considerable number of people" involved in other ancillary industries.

Transportation and fishing on the lake mainly involves use of poled dugout canoes and plank boats (Landes and Otte, 1983). The number of dugout canoes has varied significantly; when there is water in the lake, the numbers vary from as few as 100 to more than 4,000 (Fig. 4; GoM, 2005). Plank boats similarly varied in number, but gradually increased through the early 1990s (Fig. 4). This trend was interrupted by the major recession of 1995–1996 but quickly regained momentum to a peak in 2000 when more than 4,000 plank boats fished the lake. Only a small proportion of these plank boats were motorized (Fig. 4). Since the high catches and high fishing effort of the late 1990s, fishing effort, as indicated by the number of fishers, plank boats and canoes declined quickly to 2002–2003, after which no data are available.

Despite increased fishing effort, there is no evidence of major negative impacts of fishing on catches and life history traits of *Barbus* species in the lake. A recent study by Macuiane et al. (2009) found that the maximum length of *Barbus* was similar to those reported by earlier studies (Furse et al., 1979b; Bell-Cross and Minshull, 1988; Msiska, 2001; Njaya, 2001; Delaney et al., 2006). The recent persistent use of small mesh nets in Lake Chilwa makes the escape of small individuals

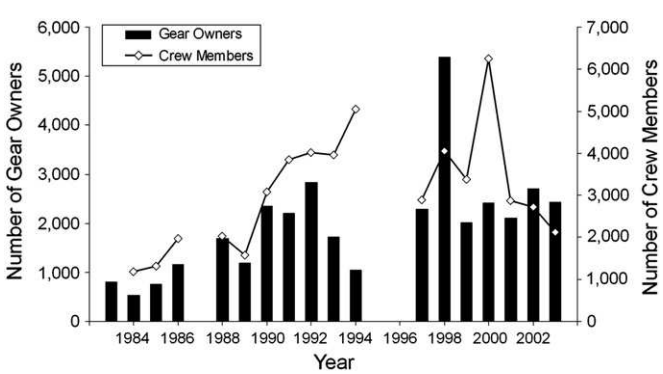


**Fig. 3.** Trends in counts of fishers from 1983 to 2002 separated by gear owners and crew members. The lake dried completely in 1995 and 1996.
Source: GoM (2002a,b).

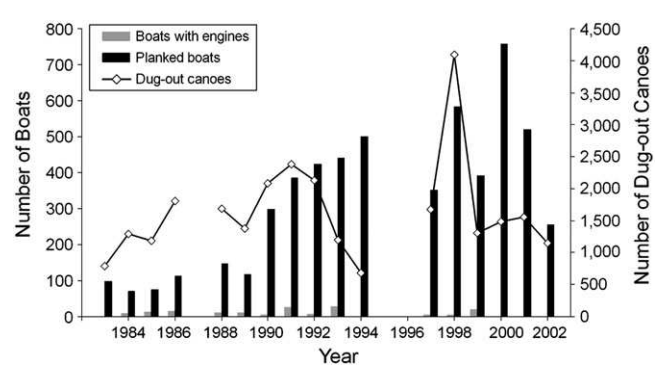


**Fig. 4.** Trends in counts of fishing craft (plank boats and dugout canoes). The lake dried completely in 1995 and 1996.
Source: GoM (2005).

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difficult. This practice might erode the resilience of *Barbus* populations by preventing small fish from reaching a size at which they can reproduce at least once before capture (King, 1995; Macuiane et al., 2009). Clearly, the relationship between fishing effort, catches and catch rates is complicated by the types of gear used (passive or active) and amount of water in the lake. Gaps in data since the peak in effort in 2000 make conclusions about the impact of declining catches and effort difficult.

The resilience of the Lake Chilwa fishery may be attributed to the breeding behaviour of *Barbus* spp, which breed in the marshes and river mouths several times per year (see Fig. 2 for collapse and recovery). In a dynamic environment such as Lake Chilwa (Msiska, 2001), breeding on more than one occasion contributes to the reproductive success of *Barbus*. Although it appears to be resilient to heavy fishing pressure, environmental degradation may pose a greater threat to sustainability of the fishery and hence livelihoods dependent on the lake. For example, Jamu et al. (2003) and Delaney et al. (2006) showed that high river sediment yield and low river flow had negative impacts on recruitment and spawning migration of *Barbus* to influent rivers. Results also showed that degraded rivers with reduced river discharge on the Malawi side were devoid of *Labeo mesops*, while the Mnembo River, which has significantly higher river discharge and a relatively undisturbed catchment, has higher fish biodiversity, including *Labeo mesops*. Recent studies (Jamu et al., 2003; Delaney et al., 2006; Macuiane et al., 2009) support results of earlier work (Furse et al., 1979b; Kalk et al., 1979) which showed that marshes serve a critical function as breeding and nursery areas and swamps as *de facto* closed areas or sanctuaries. However, burning of marshes, siltation and conversion of these marshes to rice fields threaten the sustainability of the fishery.

## Climate change in the Chilwa Basin

Exactly how climate change will affect Malawi is subject to debate, but what is not disputed is that greater variability and change in temperature and rainfall can be expected. Some studies suggest that surface air temperature in the Lake Chilwa Basin will increase by 2.6–4.7 °C by 2075 (Chavula, 1999). Precipitation trends are more contested with predicted increases of up to 8.3% in some models countered by predictions of decreases of 7% using others (Chavula, 1999). A recent UNDP climate change profile for Malawi predicts higher temperatures and suggests rainfall will remain similar but will fall in a shorter period of time with some very heavy rainfall events (C. McSweeney, M. New, and G. Lizcano, unpublished report). Extreme climatic events are not new to Malawi; the most serious being dry spells, seasonal droughts, intense rainfall, riverine floods and flash floods (Crossley et al., 1984; Nicholson, 1998; EAD, 2000). Flooding in Phalombe in 1991 reportedly resulted in more than five hundred deaths and many displaced people (http://www.africa.upenn.edu/Newsletters/mno40.html). Flooding occurred again in 2001 causing displacement and the need to bring in food relief. Such extreme events may become more common with increased climate change.

Within Chilwa basin, there is considerable spatial and temporal fluctuation in rainfall and variation in temperature. In the highland areas, rainfall ranges between 1100 and 1600 mm per year while the plains receive between 800 and 900 mm per year (EAD, 2000). Temperatures in the highlands remain fairly cool while the lower altitude zones can reach highs of 30 °C. Meteorological data for more than 30 years of record indicate rainfall varies significantly among years, with a dramatic fall in rainfall in the decade after 1985 (Fig. 5). Overall, there is no statistically significant pattern in rainfall over the period of the data set. Similarly, there are no patterns in mean or minimum temperature in the basin over the 27 years of recorded data (Fig. 6; EAD, 2000).

Climate events such as droughts or floods cause food insecurity and famine and increased pressure on natural resources such as forests and fisheries (Allison et al., 2007, 2009; Conway et al., 2005; Conway, 2009). These resources are important sources of livelihoods and become critical

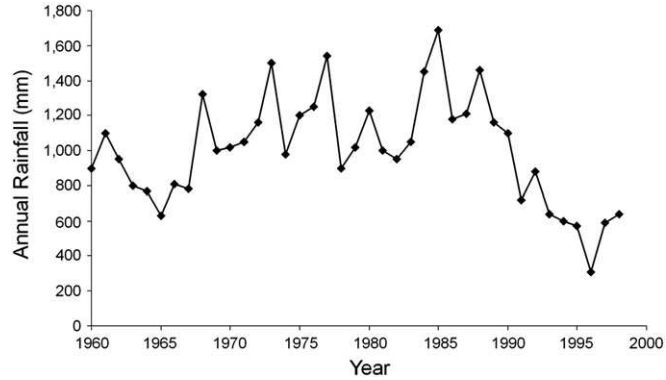


**Fig. 5.** Historical (1949–2000) mean annual rainfall recorded at stations (n = 6) in the Chilwa basin.
Source: EAD (2000).

safety nets for the poor in times of crises. Deforestation compounds and increases the intensity of flooding, which further results in the siltation of rivers, jeopardizing downstream livelihoods in floodplains (Calder et al., 1995; Mkanda, 2002; Wang et al., 2006). In the past, sector-specific strategies have aimed at solving problems of food insecurity, overfishing, deforestation, soil erosion and siltation of rivers. However, linkages between different sector strategies have been weak and have failed to address the wider impacts of climate variability and change on food production and the environment.

People living in the basin have developed specific strategies to adapt to climatic events and uncertainty. They shift homes to higher ground to avoid floods; they gather wild fruits and vegetables and hunt animals in times of crop scarcity. They use traditional medicines to cure disease, and they seek off-farm wage employment. They experiment with crop varieties and diversify their food sources as best they can. However, with increased population, fewer opportunities for off-farm work, lack of access to new and perhaps improved crop varieties, poor social services, and potentially more extreme weather conditions, the ability to adapt to climate effects is increasingly constrained. The livelihoods of most residents of the Lake Chilwa basin are already precarious and changes in climate will only increase the negative impacts on agriculture and natural resources and thus intensify residents' vulnerability.

In order to maintain the important food security, economic and social benefits of agricultural and natural resources in the face of known climate variability and anticipated climate change, planned adaptation and cross-sectoral collaboration at local, basin and national scales is required. Enhancing local-level adaptive capacity and building resilience of communities to climate variability and change must focus on strategies that lead to livelihoods' improvements. Livelihoods can be improved by strengthening existing activities (i.e. farming, livestock

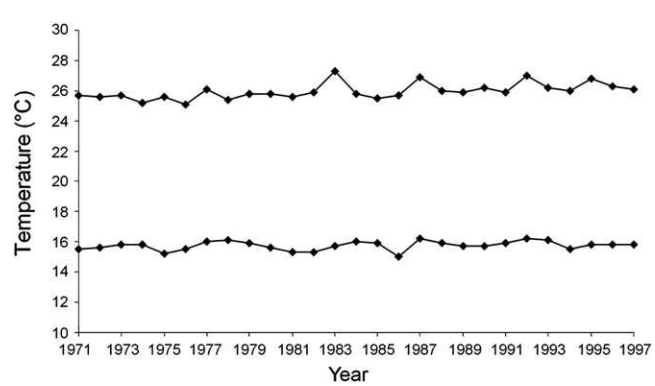


**Fig. 6.** Historical (1949–2000) mean annual minimum and maximum temperatures recorded at Makoka Agriculture Research Station.
Source: EAD (2000).

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rearing, fishing, sustainable harvesting of forest products; trading), improving access to alternative ones, strengthening household asset profiles (including social and human capital) or by addressing the institutional causes of vulnerability and insecurity. A key point is that there is scope for improving livelihoods even in contexts where natural resource dependence is high and few alternative income-generating activities are possible. Such scope is, however, very limited when the natural resource base is declining.

## Managing Lake Chilwa for resilience—problems and solutions

Climate changes, and the accompanying changes in land use in the basin, pose the greatest potential threat to the Lake Chilwa fishery. According to Moss (1979), "The Chilwa fishes are clearly well fitted to persist in the unpredictable Chilwa ecosystem, provided the refugium of swamps and streams is maintained". Moss (1979) cautions that threats to the swamps through reclamation for agriculture or perhaps as irrigation reservoirs, siltation through changes in catchment land-management, and pesticides are more dangerous than fishing. As Phipps (1973: 38) observes, the consequences of recession in Lake Chilwa waters and reduced catches are much wider than on fishing alone and "*the whole of the Chilwa plain and lake must be seen as an economic network*" as well as an ecological system. Not only are there links between fishing and various ancillary services, but, as discussed earlier, there are also complementary flows of income between fishing and farming. The integrated small-scale economy of farming, fishing and cattle-rearing of the 1960s described in Kalk et al. (1979) has now changed to one in which cattle-rearing has decreased as a saving and livelihood option, potentially decreasing the resilience of the overall livelihood system.

Around the lake, there are large-scale shifts from fishing to farming, pastoralism and other occupations when the lake dries out (and back to fishing when it refills). Phipps found in her study of "big" fishermen in 1970, that the wealthiest fishers were also those who had the greatest diversity of income generating activities. They had invested their earnings from fishing in obtaining and farming additional acreage, cultivating crops for sale, opening small businesses and investing in cattle. Such strategies highlight the importance of enhancing or maintaining the flexibility of lakeshore livelihoods rather than constraining them with fixed fisheries access arrangements (Allison and Mvula, 2002). In Lake Chilwa, where fishing has had little demonstrable impact on the capacity of fish populations to respond to the changing ecology of the basin, and where productivity is closely linked with climate, it is not useful to talk about sustainable yields in a conventional fisheries sense. As a consequence, the usual suite of management tools that seek to limit access and otherwise constrain fishing effort as the primary management response are less likely to be successful (van Zwieten and Njaya, 2003; Hara, 2006).

Malawi's fisheries have been undergoing a transition from state-based management in the colonial and post-colonial era to an arrangement known as 'co-management', where communities themselves (loosely defined as geographical entities) manage access to fisheries and enforce fishery regulations, in partnership with local and national government (Allison et al., 2002; Njaya, 2002). While many assert that co-management is the solution to many fisheries, it also has its problems. These include: power struggles between migrant and resident fishers, and between local elites and the poor; the use of new fishing regulations as a political weapon to exclude access by other ethnic groups; and a decline in the potentially beneficial role of the state in regulating the use of destructive fishing methods (Hara et al., 2002). However, co-management arrangements are also not fixed in time and change according to a wide variety of influences.

Co-management of fisheries in Lake Chilwa has evolved since it was originally implemented in 1995. The original experiment in greater community participation in management was unsuccessful because it failed to adhere to the following principles: "1) Natural resources should be managed by the people who use them; 2) At the local level, community-based natural resources management activities should be managed by democratically elected institutions or committees; and 3) The user group must be legally registered" (Lake Chilwa and Mpoto Lagoon Fisheries Management Plan Draft, 2009:2). What this disregard meant in Chilwa, was that fishers and fish traders were not well represented on the committees, which were dominated by people who were not primary stakeholders in the fishery. However, in 2000 with the passage of the Fisheries Conservation and Management (Local Community Participation) Rules of the Fisheries Conservation and Management Act (1997), elections were held and "*the rightful people elected by secret ballot to be in the Association Committee/Beach Village Committee Sub-Committee*" (ibid). While migrants may sometimes be less represented in Beach Village Committees (BVC) there is nothing to exclude them from membership. Indeed migrants register when they come into the area and get permission from the Traditional Authority to fish. These people have joined BVCs around Lake Chilwa in significant numbers (J. Wilson, personal observation).

A recent survey of the Lake Chilwa Association and its BVCs (J. Wilson unpublished report) indicates that the Association had an income of 2,160,000 Malawi Kwacha for 2008, with 37% of that income derived from fines. The rest came from the issuing of licenses. The income was then used to support by-law enforcement, pay tribute to the Traditional Authority, convene meetings and to assist sick members of the associations. Occasionally, money was also used to help with funeral expenses. Thus, income from BVC fines and licenses provides important social services for its members in addition to sustaining the committees (ibid). Members of the BVC themselves have identified what they perceive to be threats to the important fish species found in the lake. These threats include: cutting of lake plants where fish escape and hide; using gauze wire around plant beds; using seine nets in January, February and March; using poison in rivers and pools; catching fish using gauze wire and mosquito nets; and using fish traps in the rivers. They proposed a complete ban on these methods to ensure the health of the fishery.

Measures suggested by the BVCs to conserve fish stocks in Lake Chilwa, such as closed seasons, point to an awareness of the importance of the wider environment in the health of the fishery. Various land-use practices in the basin jeopardize fish breeding grounds, the surrounding wetland functions and broader ecosystem integrity. This broader land-use context is important to consider in plans for fisheries management. Poverty and pressure on agricultural land and need for energy (e.g. fuel wood) drives many of the practices which are degrading the basin's environment, from deforestation to increased river bank cultivation. Both these practices lead to increased siltation which causes heavy silt loads in the rivers and ultimately deposits soil in the floodplains. Rivers, with their increased silt deposits, flood more easily and water disperses into villages (causing great destruction), and floodplains. This flooding may result in an overall decrease in the volume of water reaching the lake.

Where rivers flow through the swamps there are always aquatic habitats of some kind that persist even during low lake level periods. This is a vital characteristic that affects the productivity of the lake fishery because at least two of the important fish species use the swamps as breeding sites. At low lake levels, perennial swamp habitat is key. Heavy fishing pressure in these reduced aquatic habitats was suggested as one reason for the longer than expected recovery of Makumba after the 1967–1968 dry period and it is inevitable that this will occur again affecting all three commercial fish species because of increasing human populations on the plains and because of the conversion of perennial swamp habitat to rice cultivation. At a catchment level it is crucial to protect the sources of perennial water flow in the upper catchments and to protect the downstream lake edge marsh and swamp habitats associated with these perennial flows.

The wider environmental and socio-economic context, rather than the state of fish stocks, has affected fisheries management in many

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places in Africa. In Jul-Larsen's (2003) study, access to the fishery by migrants was denied not because of any concerns over the state of fish stocks, but rather the desire to keep migrants from making claims to land and settling in villages around Lake Kariba. In Lake Chiuta, close to Chilwa, devolution of control to the Fisheries Association has been successful in regulating against destructive fishing practices. The Association kept migrants out of the fishery not because they were outsiders but rather because they refused to abide by the by-laws of the association and were thus threatening the fishery resources. The Association has put a ban on the gear types used in Lake Malombe to which many attribute the collapse of that fishery.

The challenge in managing for resilience is how to address some of the wider environmental and socio-economic conditions that affect the lake. If poverty increases and crops fail due to climate conditions, there will most certainly be greater pressure on the lake, particularly through encroachment into the wetlands. If the lake dries again due to extended periods of drought or if water levels decline significantly due to less water flowing into the Lake because of upstream practices, the livelihoods of a vast number of Chilwa residents will be threatened.

## Concluding remarks

Greater understanding of how people cope with, and adapt to, fisheries with extreme natural variations would assist in developing adaptation strategies to climate change. The relative risk of climate change on fisheries sectors also needs to be understood in the context of impacts on other natural resource sectors and on other hazards that result in high levels of poverty, including food insecurity, epidemic disease, conflict, political marginalization, inequity and poor governance. In the context of small scale fisheries, adaptations must focus on building institutions and rules of management that will increase the capacity of ecosystems and people to accommodate unpredictable change. Flexible and reactive institutions will offer the best chances of minimizing the effects of irreversible change. While co-management reform has allowed greater decision making over their management by the actual resource users, the Lake Chilwa Fisheries Association and the BVCs are still focused primarily on the lake's resources. Linking these governance units with resource users upstream could increase awareness of inter-linkages of land use practices across scales and lead to improved management of the entirety of the basin's resources. Ultimately, management of the Lake Chilwa fishery needs to take into account the wide variety of resource use across the basin and the socio-economic context in which this use takes place.

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